

Constructors in Java

A **constructor** in Java is a special method used to initialize objects. It is called automatically when you create an object using the **new** keyword.

Note-> constructor as name suggests should be used to construct that is how people get confused, Constructor is only used for initialization and will be called only if the object is already created using new keyword.

Key Characteristics Constructor has

1. the **same name as the class**
2. **No return type** (not even **void**)
3. Access modifier can be used(public,protected,private,default)
4. **No Non access modifiers.**(static,final,volatile,abstract,synchronised,transient,native,strictfp)
5. Used to initialize instance variables

Types of constructor

1. **Default Constructor**
2. **No-Argument Constructor (User-defined)**
3. **Parameterized Constructor**

Default Constructor

If we do not create any constructor,jvm will create a Default constructor which will be called to initialize instance variable with default values.

```
class Car {  
  
    int wheels; //instance variables  
  
    String brand;//instance variables  
  
    void show(){  
  
        System.out.println("Brand": " + brand);  
    }  
}
```

```
        System.out.println("Wheels": " + wheels);
    }

    public class Main {

        public static void main(String[] args) {

            Car car1 = new Car(); // Calls default constructor

            car1.show();

        }

    }
```

No-Arg Constructor

A constructor that takes no parameter(arguments) defined by user.

```
class Car {

    int wheels;

    String brand;

    // No arg Constructor

    Car() {

        wheels = 4;

        brand = "Unknown";

    }

    void show() {

        System.out.println("Brand": " + brand);

    }

}
```

```

        System.out.println("Wheels": " + wheels);
    }
}

public class Main {

    public static void main(String[] args) {

        Car car1 = new Car(); // Calls No -arg constructor

        car1.show();

    }

}

```

Parameterised Constructor

A constructor that takes parameter(arguments) defined by user,Used to initialize objects with specific values.

```

class Car {

    int wheels;

    String brand;

    // No arg Constructor

    Car(int a , String b) {

        wheels = a;

        brand = b;

    }

}

```

```

    void show() {

        System.out.println("Brand": " + brand);

        System.out.println("Wheels": " + wheels);

    }

}

public class Main {

    public static void main(String[] args) {

        Car car1 = new Car(4,"TATA"); // Calls No -arg constructor

        car1.show();

    }

}

```

this Keyword in Constructors

Used to refer to the current object.

```

class Car {

    int wheels;

    String brand;

    // No arg Constructor

    Car(int wheels , String brand) {

        wheels = this.wheels;

        brand = this.brand; // differentiates instance variable from parameter to
the compiler

    }

}

```

We will further study about Constructor once we study Inheritance.